

A mechanism for slow release of biomagnified cyanobacterial neurotoxins and neurodegenerative disease in Guam



As root symbionts of cycad trees, cyanobacteria of the genus *Nostoc* produce β -methylamino-L-alanine (BMAA), a neurotoxic nonprotein amino acid. The biomagnification of BMAA through the Guam ecosystem fits a classic triangle of increasing concentrations of toxic compounds up the food chain. However, because BMAA is polar and nonlipophilic, a mechanism for its biomagnification through increasing trophic levels has been unclear. We report that BMAA occurs not only as a free amino acid in the Guam ecosystem but also can be released from a bound form by acid hydrolysis. After first removing free amino acids from tissue samples of various trophic levels (cyanobacteria, root symbioses, cycad seeds, cycad flour, flying foxes eaten by the Chamorro people, and brain tissues of Chamorros who died from amyotrophic lateral sclerosis/Parkinsonism dementia complex), we then hydrolyzed the remaining fraction and found BMAA concentrations increased 10- to 240-fold. This bound form of BMAA may function as an endogenous neurotoxic reservoir, accumulating and being transported between trophic levels and subsequently being released during digestion and protein metabolism. Within brain tissues, the endogenous neurotoxic reservoir can slowly release free BMAA, thereby causing incipient and recurrent neurological damage over years or even decades, which may explain the observed long latency period for neurological disease onset among the Chamorro people. The presence of BMAA in brain tissues from Canadian patients who died of Alzheimer's disease suggests that exposure to cyanobacterial neurotoxins occurs outside of Guam.